Aman Jayesh

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Education

International Institute of Information Technology, Hyderabad **Telangana**
Bachelor of Technology in Computer Science and Engineering *July 2024 – May 2028*

- **Relevant Coursework**: Introduction to Information Security, Operating Systems and Networks, Data Structures and Algorithms, Design and Analysis of Software Systems, Probability & Statistics, Software Programming for Performance.

Experience

Undergraduate Researcher at CSTAR under Prof. Ankit Gangwal **March 2026 – Present**

- Investigating how backdoor poisoning attacks and adversarial examples can compromise global model integrity in federated learning systems, as part of ongoing ML security research at CSTAR.

Team Head at Open Source Developers Group, IIITH **March 2026 – Present**

- Headed design and operations for IIITH's open source community, led end-to-end event logistics for 150+ attendee events, coordinated across organizer teams, and oversaw visual branding and merchandise.

Project Intern at SleepAlfa **Jan 2026 – April 2026**

- Worked with a team of 7 at a startup to develop a web application to make sleep assessments more accessible for patients and easier to evaluate for doctors.
- Architected webhook integrations (YouForm, Cal.com) and was the sole backend engineer on deployment pipeline (Vercel + Render)

Research

SAMSTeG: Noise-Robust Keyword Spotting | *Deep Learning, TinyML* [GitHub] CVEST Lab IIITH

- Authored a research paper for IEEE Embedded Systems Letters Journal (still under review) detailing a novel noise-robust keyword spotting architecture for extreme edge devices under Prof. Abhishek Srivastava.
- Proposed Multi-Scale Temporal Gating (MSTG) and Hybrid Dropout and achieved 95.71% accuracy on Google Speech Commands dataset, matching SOTA models like BC-ResNet.

OSCAPegoat | *Python, LLMs* [GitHub] CyberMANTHAN Lab IIITH

- Achieved 70% automated STIG compliance remediation by benchmarking 5 local LLMs against OpenSCAP failing tests at CyberManthan IIITH.
- Built error-resilient scripts with a 3-iteration self-correcting loop to enable fully automated compliance remediation.

Projects

Felicity: College Fest Portal | *React, Node.js, MongoDB* [GitHub] DASS Course Project

- Built a full-stack event management portal for college fests with a three-tier RBAC system (Admin, Organizer, Participant) enforced via JWT authentication middleware on the Express backend and protected routes on the React frontend.
- Implemented event creation with capacity limits, participant registration, and role-aware dashboards, with the frontend deployed on Vercel and a MongoDB-backed REST API.

OpenWheel-Gnome | *Linux, C++* [GitHub] Open Source Project

- Extended a Linux userspace HID controller for the ASUS ProArt dial to support GNOME, with an architecture spanning a C daemon reading raw device events and a C++/Qt frontend using D-Bus IPC and profile-based input simulation.

Achievements & Extracurriculars

- 2x Hackathon Winner @IIITH (Rs. 35000 Winnings): Build2Break 2025 1st Place, Megathon 2024 2nd Place
- Former Kerala State Team Chess Champion.
- JEE: Mains 1261 and JEE Advanced 2341 CRL

Technical Skills

Languages: C, C++, Python, Bash, SQL
Frameworks: TensorFlow, PyTorch, Scikit-Learn, NumPy, Matplotlib, Librosa, Node.js, Express, React
Tools & Platforms: Git, GitHub, Shodan, ESP32, Edge Impulse, HuggingFace, Ollama
Certifications: Machine Learning Specialization (DeepLearning.AI – Stanford)